

String Matching Review

Questions and Answers

Indexing Convention

For algorithm traces, text and pattern positions are shown with 0-based indexing.

Question 1

List the prefixes of the string $P = \text{aaabbaaa}$ that are also suffixes of P .

Answer 1

The prefixes that are also suffixes are

`a, aa, aaa, aaabbaaa.`

If only proper prefixes are required, remove the whole string:

`a, aa, aaa.`

Question 2

What is the longest proper prefix of `cgtacgttcgtacg` that is also a suffix?

Answer 2

The longest proper prefix-suffix is

`cgtacg.`

It appears at both ends:

`cgtacgttcgtacg.`

Question 3

Draw a figure illustrating the comparisons done by brute-force pattern matching for text `aaabaadaabaaa` and pattern `aabaaa`.

Answer 3

Let

$T = \text{aaabaadaabaaa}, \quad P = \text{aabaaa}.$

Shift	Comparison result	Outcome
0	$a = a, a = a, a \neq b$	fail
1	$a = a, a = a, b = b, a = a, a = a, d \neq a$	fail
2	$a = a, b \neq a$	fail
3	$b \neq a$	fail
4	$a = a, a = a, d \neq b$	fail
5	$a = a, d \neq a$	fail
6	$d \neq a$	fail
7	$a = a, a = a, b = b, a = a, a = a, a = a$	match

Therefore the pattern occurs at

shift 7.

The brute-force method performs 24 character comparisons.

Question 4

Repeat the previous matching example using the Knuth-Morris-Pratt algorithm, not counting the comparisons made to compute the failure function.

Answer 4

For $P = \text{aabaaa}$, the KMP prefix/failure table is

$$\pi = [0, 1, 0, 1, 2, 2].$$

Text index	$T[i]$	Main action	Matched length after step
0	a	extend	1
1	a	extend	2
2	a	fallback $2 \rightarrow 1$, then extend	2
3	b	extend	3
4	a	extend	4
5	a	extend	5
6	d	fallback $5 \rightarrow 2 \rightarrow 1 \rightarrow 0$	0
7	a	extend	1
8	a	extend	2
9	b	extend	3
10	a	extend	4
11	a	extend	5
12	a	extend to full match	2

The full match ends at text index 12, so the shift is

$$12 - 6 + 1 = 7.$$

KMP reports the pattern at shift 7.

Question 5

Compute a table representing the KMP failure function for the pattern string **cgtacgttcgtac**.

Answer 5

For $P = \text{cgtacgttcgtac}$, the prefix/failure function is:

q	0	1	2	3	4	5	6	7	8	9	10	11	12
$P[q]$	c	g	t	a	c	g	t	t	c	g	t	a	c
$\pi[q]$	0	0	0	0	1	2	3	0	1	2	3	4	5

$$\pi = [0, 0, 0, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5].$$